CAT-2019
Slot-2
EXPLANATORY
ANSWERS

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SECTION-I
VERBAL ABILITY & READING COMPREHENSION

Q.1) The term 'digital colonialism' appears in the 1st paragraph: *But the project is raising questions about Google's motivations and about who should own the digital copyrights. Some critics call it a form of "digital colonialism."* The paragraphs that follow elaborate on these issues, i.e. Google's motivations and the ownership of digital copyrights. Option 2 specifies the latter, i.e. critics' disagreement about copyright ownership: *CyArk owns the copyrights of the scans — not the countries where these sites are located. That means the countries need CyArk's permission to use these images for commercial purposes.* [Paragraph 5] Retain option 2. The passage doesn't account for options 1 and 4; refer to the last paragraph: *The company says it works closely with authorities during the process, even training local people to help.* Because the passage doesn't doubt this claim, both options can be rejected. Option 4 is incorrect; nowhere does the passage suggest that the scanning process damages the cultural sites. Reject option 3. Hence, [2]

Q.2) Paragraph 4 introduces Ethan Watrall: *… says this project is just a way for Google to promote Google … Watrall says these images belong on the site of a museum or educational institution, where there is serious scholarship and a very different mission* … He is against the project where Google and CyArk have collaborated to create 3D scans of historic sites. The passage refers to 'the world's most precious cultural sites', not just those in the developing countries; besides, he is not against 'the use of digital technology' in such historic sites, as option 1 suggests. Reject option 1. As you can see in paragraph 4, Dr. Watrall's concerns are different from what option 2 states—it's more about how Google will market itself through the images, instead of putting them to more productive use. Reject option 2. We don't have enough data in support of option 3. Option 4 is the best choice as paragraph 4 clearly states that Dr. Watrall is 'not comfortable with the arrangement between CyArk and Google'. Hence, [4].

Q.3) The question requires us to choose that option which strongly invalidates Dr. Watrall's arguments. Dr. Watrall is against the project where Google and CyArk have collaborated to create 3D scans of historic sites, because he doubts Google's motivation-- he believes it to be rooted in personal gain. Besides, he believes that a more deserving setting for these images is either a museum/ an educational institution. Copyright issues are brought in later in the passage; Watrall only hints at this. That CyArk does not own the copyright on these images doesn't weaken Watrall major concern, which is that the scanned images are not put to productive use. Reject option 1. That CyArk is banned in 'certain countries' do not, in any way, challenge Dr. Watrall's argument. Reject option 2. Ads are only an instance of how Google could benefit from the scanned images; option 4 doesn't invalidate Dr. Watrall's argument as Google could benefit from the scanned images even without the aid of ads. Option 3, on the other hand, sets right two of Watrall's objections. Hence, [3].

Q.4) Refer to the 6th paragraph: *Erin Thompson, a professor of art crime at John Jay College of Criminal Justice in New York City, says it's the latest example of a Western nation appropriating a foreign culture, a centuries-long battle…* In the context, the word *appropriate* means 'to take something from a culture and use it as part of your own'. Dr. Thompson considers the project to be the latest example of cultural appropriation. Only option 2 presents a comparable scenario — a museum showcasing an exhibit that doesn't belong to or represent its culture. Reject all other options. Hence, [2]

Q.5) Three out of the four options support the initiative by Google. We are required to choose the exception. Option 1 clearly supports the project: *The digital renditions allow viewers to virtually wander the halls of the temple, look up-close at paintings and turn the building over, to look up at its chambers…* [Paragraph 2] Reject it. Consider option 2; here, the word *project* [to represent oneself in a particular way] reflects the fears of the critics: *… collaboration could be an attempt by a large corporation to wrap itself in the sheen of culture.* [Paragraph 4] Retain option 2. Option 3 challenges the claim of those like Dr. Watrall, who believes that the motivation for the project is rooted in Google's personal gain. Reject option 3. Option 4 can also be rejected; refer to paragraph 2: *Before the earthquake at Bagan, many of the temples on the site were scanned. . . . [These] scans . . . are on Google's Arts & Culture site.* Hence, [2]

Q.6) We are required to choose that option which invalidates the author's argument. The term *orientalism* refers to 'the representation of Asia in a stereotyped way that is regarded as embodying a colonialist attitude'. Whether or not Egyptians practice orientalism doesn't influence the author's claims—he only defends himself by presenting a scenario, in which a common tongue erases the cultural barriers. Reject option 1. Options 2 and 4 support the author's argument. Refer to paragraph 3: *And both the Chinese and the Egyptians welcomed me because I spoke their languages. My identity as a white male was far less important than my ability to communicate.* The same paragraph also states that '… empathy and understanding are not inherited traits, and they are not strictly tied to gender and race'. On the other hand, option 3 states the inverse of this, thus challenging the author's argument. Hence, [3].

Q.7) Refer to the last paragraph: *If all of us now stand beside the same river, speaking in ways we all understand, who's looking east and who's looking west? Which way is Oriental?* Through these words, the author asserts his belief that identity linguistics doesn't exist. Option 1 presents a comparable statement. None of the other options bring in the relevant idea, i.e. 'how speaking a common language enables us to overcome the cultural barriers'. Reject them. Hence, [1]

Q.8) According to the author, 'you can always learn another language and change who you are.' [Last Paragraph] The author's experience with the Chinese and the Egyptians, who welcomed him because he spoke their language, points to how 'the words that run throughout your mind are at least as fundamental to your selfhood as is your ethnicity or your gender.' [Paragraph 3] Naturally, the author would encourage the French ethnographer [an *ethnographer* is 'someone who studies and describes the culture of a particular society or group'], but also ask her to learn the language of that culture she is planning to study. This is what option 1 says. Retain it. Option 2 is incorrect; it not only says nothing about the need to learn the language of the Nigerian tribe, but also contradicts the author's viewpoint by focusing on 'racial and gender identities'. Reject it. Options 3 and 4 totally reject the author's belief that 'learning the language of a foreign culture gains one easy access to that culture'. Reject them. Hence, [1].

Q.9) The question requires us to choose an option that is NOT responsible for the transformative effect that a language has on its learner. Refer to paragraph 3: *My identity as a white male was far less important than my ability to communicate.* Option 1 reflects this. Reject it. Refer to paragraph 3 again: *An individual who wrestles with a difficult language can learn to be more sympathetic to outsiders and open to different experiences of the world...* Thus, option 2 is also a factor that leads to language-learners' transformation. Reject it. 3 is a tricky option; it could be understood in two ways— either that language breaks cultural barriers or that it reinforces them. The former understanding is in tune with the passage: *…you can always learn another language and change who you are.* [Paragraph 4] Retain it. The

author doesn't bring in the topic 'evolution of language' [Option 4] in the passage. Therefore, option 3 can be rejected as option 4 is the exception we are looking for. Hence, [4]

Q.10) The excerpt given in the question appears in paragraph 2: *… because modernity was externally introduced, it is explanatorily unhelpful to apply the logical format of the 'transition process' to this pattern of change … the logical structure of endogenous change does not apply here. Here transformation agendas attack as an external force.* This part explains why the British initiative to introduce the logic of modernity into Indian society came to assume an external character. One reason is that it was not endogenous; something that is *endogenous* 'comes from within something, for example a system or a person's body or mind'. It was an external force rather than a national movement-- it was more forced than natural. Reject option 1; *endogamy*, which refers to 'the practice of marrying within a specific social group' and *exogamy* are not discussed in the passage. Option 2 is self-contradictory; colonialism, as it's understood commonly, can never be endogenous; in most cases, it is an external force. Besides, the passage doesn't talk about the 'endogenous logic of colonialism'. Reject option 2. Option 3 is a valid conclusion based on the given statement. Retain it. Option 4 is incorrect; *logical structure* refers to the way information is organized, while *logic*, as used in option 4, refers to 'reason'. Besides, the passage doesn't endorse the need to transform colonized societies by means of external force. Reject it. Hence, [3].

Q.11) The excerpt given in the question appears in paragraph 2: *For, whatever their sense of the strangeness of the country and the thinness of colonial presence, the British colonial state represented the great conquering discourse of Enlightenment rationalism … As inheritors and representatives of this discourse, which carried everything before it, this colonial state could hardly adopt for long such a self-denying attitude. It had restructured everything in Europe … Consequently, the colonial state could not settle simply for eminence at the cost of its marginality; it began to take initiatives to introduce the logic of modernity into Indian society.* As per this, the British would remain at the margin as long as it maintains its self-denying attitude. Here, 'the self-denying attitude' refers to the 'mix of military power and subtle diplomacy' that the British employed on India initially; in order to overcome this *marginality* ['being outside of the mainstream of something'], they introduced their own logic of modernity into Indian society. Simply put, in order to gain access to the culture, they moulded the Indian society to be like them. Options 1 and 3 can be rejected as they don't clarify the concept of *marginality*. Option 4 is incorrect—as per the passage, *the colonial state could not settle simply for eminence at the cost of its marginality*. Option 2 rightly captures the sense of the given statement. Here, the British modernity stands for their eminence [or the discourse of Enlightenment rationalism characteristic of them]. Hence, [2].

Q.12) The passage consists of two paragraphs. The first paragraph starts by describing the two phases that the British **colonial policy** went through, in India—"At first, the new colonial apparatus exercised caution, and occupied India by a mix of military power and subtle diplomacy… [then] it began to take initiatives to introduce the logic of modernity into Indian society." [Paragraph 1] The British philosophy of ***enlightenment*** *rationalism* compelled them to change the self-denying attitude that they embraced during the initial stage of colonisation; this was what caused the shift in the colonial strategy. The second paragraph describes the after-effect of this 'introduction of **external modernity**'— Indians came to associate it with **subjection**, and India witnessed *the creation of a degenerate version of capitalism — what early dependency theorists called the 'development of* ***underdevelopment'****.* [Paragraph 3] The words in bold correctly represents the flow of the arguments in the passage. Hence, [3]

Q.13) We are required to choose that option which contradicts the arguments given in the passage. The question is easy to solve if the passage is clear. As per option 1, the British shifted to the second phase of colonial policy because Indians resisted modernity. This is false; 'introduction of external modernity' represents the second phase of the British colonial policy: —"At first, the new colonial apparatus exercised caution, and occupied India by a mix of military power and subtle diplomacy… [then] it began to take initiatives to introduce the logic of modernity into Indian society." [Paragraph 1] Retain option 1. Option 2 is true as per the last paragraph: "Economic reforms, or rather alterations . . . did not foreshadow the construction of a classical capitalist economy … What happened was the creation of a degenerate version of capitalism —what early dependency theorists called the 'development of underdevelopment'." Reject option 2. Option 3 is also true as per the passage: "It had restructured everything in Europe … and would do the same in India, particularly as some empirically inclined theorists of that generation considered the colonies a massive laboratory of utilitarian or other theoretical experiments." [Paragraph 1] Reject option 3. Option 4 is also true as per the passage: "… what was proposed to be built was something like European capitalism … in any case, historians have forcefully argued that what it was to replace was not like feudalism …" [Paragraph 2] Here, the author describes the limitations of the 'transition process' that the British undertook in India. Hence, [1].

Q.15) The word appears in paragraph 6: "The dilemma is obvious. Pick small, poor towns … it is hard to attract the most qualified workers; opt for larger cities with infrastructure and better-qualified residents, and the country's most deprived areas see little benefit …" It describes one of the disadvantages of decentralising government jobs— the difficulty in choosing between small towns and large cities while deciding on an ideal site for the relocation of government agencies. Option 1 is almost a reiteration of the excerpt given above. Retain option 1. Option 2 is comparable to option 1; though, 1 is a better choice because the "dilemma" doesn't relate to 'the largest city' [option 2]. According to the passage …*where government jobs go, private ones follow*. However, encouraging private enterprises to relocate to smaller towns has nothing to do with the word "dilemma" as used in the passage. Reject option 3. Option 4 could be linked to one of the three specific aims of decentralising the central government: to improve the lives of both civil servants and those living in clogged capitals. However, this is not what the word "dilemma" refers to in the passage. Reject option 4. Hence, [1]

Q.16) Refer to the 4th paragraph: "After the second world war Britain moved thousands of civil servants to "agreeable English country towns" as London was rebuilt." This is an illustration of the author's claim that *the first aim—improving living conditions—has a long pedigree.* In the context, the word *pedigree* means 'the history of an idea or activity'. The author is trying to say that the practice of shifting civil servants, to improve their living conditions, has a long history. Options 1, 2 and 4 do not express this notion, but option 3 does. Hence, [3]

Q.17) The question requires us to choose the option that doesn't support the decentralisation of central government functions. Option 1 states one of the advantages of the same: "Office space costs far more in capitals … Agencies that are moved elsewhere can often recruit better workers on lower salaries …" [Paragraph 4] Reject option 1. Option 2 also states an advantage: "Wonks in the sticks will be inspired by new ideas that walled-off capitals cannot conjure up." [Paragraph 2] Reject option 2. Option 3 is least likely to be used in support of decentralisation of civil servants; as per the last paragraph, decentralisation could worsen corruption, as journalists, who tend to live in the bigger cities, become less watchful of those in power. In other words, decentralisation could break the nexus, or connection, between bureaucrats and media in the capital, and this, in turn, could make political corruption worse. Retain option 3. Option 4, like the first two options, states an advantage: "Autonomous regulators perform best far from the pressure and lobbying of the big city." [Paragraph 2] Reject it. Hence, [3]

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Q.18) We are required to choose an option that doesn't state a disadvantage of decentralisation, as far as the author is concerned. Option 1 is stated in the passage: "But swapping the capital for somewhere smaller is not always agreeable. Attrition rates can exceed 80%." [Paragraph 4] In the context, *attrition rate* refers to the rate at which employees voluntarily leave an organization. Reject option 1. Option 2 is also stated in the passage; refer the last paragraph: "… decentralisation begets corruption by making government agencies less accountable …" Reject option 2. Option 3 is stated in the 6th paragraph: "Pick small, poor towns, and areas of high unemployment get new jobs, but it is hard to attract the most qualified workers …" Reject option 3. The passage doesn't talk about pollution levels and congestion in the new locations. Therefore, option 4 is the exception. Hence, [4]

Q.19) This is a straightforward question whose answer can be spotted in the first paragraph: "In the post-colonial fervour of the 20th century, coastal capitals picked by trade-focused empires were spurned for "regionally neutral" new ones …" Here, the phrase *trade-focused empires* points to the answer, which is option 1. All of the remaining options can be rejected. Hence, [1]

Q.20) Calthorpe's statement appears in the third paragraph: "The city is the most environmentally benign form of human settlement. Each city dweller consumes less land, less energy, less water, and produces less pollution than his counterpart in settlements of lower densities." Something is *jarring* if it 'disagrees or seems wrong or unsuitable.' In other words, what Calthorpe said was against common belief or expectation. Therefore, answer to the above question would be what most people have understood about cities and their eco-friendliness. Options 1, 2 and 3 are not valid inferences as they are not specifically about eco-friendliness of the cities. Option 4 is the best inference that can be made in the context. Hence, [4]

Q.21) We are required to choose an option that states a disadvantage of living in a city. Option is stated as an advantage in the passage; in fact, it's the main topic of discussion: "… Each city dweller consumes less land, less energy, less water, and produces less pollution than his counterpart in settlements of lower densities …" [Paragraph 3] Reject option 1. Refer to the last paragraph: "… cities offer more than just jobs. They are transformative: in the slums, as well as the office towers and leafy suburbs, the progress is from hick to metropolitan to cosmopolitan …" The above excerpt states employment and transformation [or progress] as two of the reasons why cities are overall a net good for those who move there. Reject options 2 and 4. However, that cities have suburban areas as well as office areas is not stated as an advantage of city-dwellers. Hence, [3]

Q.22) Each of the 4 options describes the negative side of cities. However, we are required to choose that which weakens the author's argument regarding the greenness of cities, which is stated in the first paragraph: "… they look organic. Squatter cities are also unexpectedly green. They have maximum density … and have minimum energy and material use." In the context, *greenness* refers to '(a product or a service) that is not harmful to the environment'. Option 1 can be rejected because an increase in the cost of utilities doesn't challenge the author's argument that the consumption of utilities is less due to the compactness of cities. As per option 2, the high density of cities is causing global warming; this challenges the author's argument that maximum density makes them eco-friendly. Retain it. Options 3 and 4 can be rejected as the author acknowledges that crime and health concerns are features of fast-growing cities: "Of course, fast-growing cities are far from an unmitigated good. They concentrate crime, pollution, disease and injustice as much as business, innovation, education and entertainment…" Thus, they don't challenge the author's argument. Hence, [2].

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Q.23) Refer to the first paragraph: "They have maximum density … and have minimum energy and material use. People get around by foot, bicycle, rickshaw, or the universal shared taxi." Refer to the 2nd paragraph: "… in most slums recycling is literally a way of life. The Dharavi slum in Mumbai has 400 recycling units and 30,000 ragpickers. Six thousand tons of rubbish are sorted every day." From the above excerpts, options 1, 2 and 4 can be concluded to be reasons why the author calls squatter cities environment-friendly. However, that their streets are kept clean, is not mentioned anywhere in the passage. Hence, [3]

Q.24) Refer to the fifth paragraph: "[T]he nationally subsidised city of Manaus in northern Brazil "answers the question" of how to stop deforestation: give people decent jobs." The answer to the question is explicitly stated in the above excerpt—[Manaus] "answers the question" of **how to stop deforestation**. Therefore, the author's primary purpose is not to promote cities as employment hubs, or describe the infrastructure efficiencies in cities; neither is it to explain where cities source their labour for factories. Only option 4 states the correct reason. Hence, [4]

Q.25) The first statement states one of Socrates' teaching—*knowledge of oneself is the path to true wisdom*. Statement 2 states how the rhetorical method called *illeism* (speaking about ourselves in the third person) is beneficial— for what, we don't know yet. Statements 3 and 4 speak about rumination—how it is not the ideal way to achieve self-realization [it leads to impaired decision making and triggers depression]. Statement 5 talks about a 'change in perspective', and how it's useful in allowing us to see past our biases; this could be a reference to the idea of *illeism* [Sentence 2]. Thus, we have two mandatory pairs—2-5 and 4-3. Sentence 1, though closely related to the topic of discussion, is the best choice for the stand-alone sentence, as all other sentences closely pairs with each other. Hence, [1]

Q.26) Hybrid organizations can (better than non-hybrid organizations) mobilize supporters from allied movements. For example, those with past involvement in non-anti-war movements were more likely to join the post-9/11 U.S. anti-war movement than the rest. Hence, [4].

Q.27) Though all sentences mention *single-panel comics*, we can't yet conclude that it constitutes the main topic of discussion. Sentence 1 talks about 'a particularly interesting example of inference that occurs in many single panel comics'. Sentence 2 describes the creators' role in the making of single-panel comics. Sentences 3, 4 and 5 deal with how single-panel comics work— in order to be able to enjoy the joke, readers must imagine the preceding and succeeding scenes or, in other words, the 'missing' panels. As you can see, sentences 3, 4 and 5, together, give us an example of inference—a guess that you make based on the information that you have— through single-panel comics. Therefore, the central topic of this paragraph is not *single-panel comics*, but *inference*. Therefore, sentence 2 is the exception. Hence, [2]

Q.28) Only sentence 3 is eligible to be the starter of the paragraph; 'such a belief' in sentence 1, 'these parts' in sentence 2 and 'it is an example' in sentence 4 make each of them inadequate as a starter. Sentence 3 describes the earlier understanding of the 'balance in nature'—*it was part observational, part metaphysical, but never scientific.* Sentences 4 and 2 form a mandatory pair in that order. Sentence 4 talks about teleology-- *the notion that nature has a predetermined destiny associated with its component parts*. Sentence 2 states how these parts fit together into an 'integrated, well-ordered system that was created by design'. Sentence 1, which brings in the idea of divine influence— *a purpose presumably imposed by the goodness and wisdom of a deity*— is an apt conclusion because it accounts for the idea of predestination. Hence, 3421

Q.29) Neither sentence 3 (… perform it …) nor sentence 4 (however …) can be the starter. Sentence 1 talks about atonal music-- specifically, about how an uninitiated person perceives it. Sentence 2 defines atonality. Between sentences 1 and 2, sentence 2 is the better starter. Sentence 4 indicates a contrast; the only sentence that it can follow is sentence 1. Together, they say that though atonal music can't be enjoyed by those without the formal knowledge of it, it is one of the most important movements in 20thC music. Thus, we have the mandatory pair 1-4. The remaining sentence, i.e. 3 gives further support to the claim made in sentence 4—one requires significant amount of knowledge, skill, and technical expertise to compose or perform it. Thus, it is an apt conclusion to the paragraph. Hence, 2143

Q.30) Consider sentence 4—feminist literature has always represented clock-time as a masculine artefact representative of a 'time is money' perspective. Sentence 1 describes two adverse effects of such a representation-- clock-time is contrary to 'women's time' but synonymous with *economic rationalism*; in other words, 'women's time' is contrary to *economic rationalism*. Thus, we get the mandatory pair 4-1. Sentence 3 points out another inconsistency—(for both genders) it is found that caring time is often more focused on the clock than generally theorised. In other words, dichotomies of 'men's time', 'women's time', clock-time, and caring time are not exclusive; they overlap at different points. Thus, sentence 2 gives an apt conclusion to the paragraph. Hence, 4132

Q.31) Sentence 2 states that adaptive behaviour prefers a *cognitive economy*; the sentence also defines the term—it involves treating objects as equivalent. Sentence 4 further explains this—in a category, all things are equivalent to each other and different from things outside that category. This characterises adaptive behaviour. Thus, we get a mandatory pair, 2-4. Sentence 3 explains how the information we receive from around us is not arbitrary. This reinforces what the 2-4 pair indicates— there is an order and structure to the reception and categorization of information. Sentence 1 concludes the paragraph. Hence, 2431

Q.32) The first part of the passage asserts that language is unique to Homo sapiens. However, in the second half, it points out the limitations of this claim. Firstly, 'language' is not a monolithic entity—it's the product of a complex evolutionary process. Besides, there are other factors that characterise language (besides the uniquely linguistic features of the human language). For example, features of cognition, such as memory. Option 1 is not incorrect; however, it fails to mention the impact of cognitive features on language. Reject it. Option 2 states both—evolution of language and the impact of cognitive features in the process. Retain it. Option 3 can be rejected as it's only a rewording of the first part of the passage, which has been challenged in the second part. Option 4 is incorrect—the passage doesn't state anywhere that language is a derived trait. Reject it. Therefore, option 2 is the best choice among the four options. Hence, [2]

Q.33) The passage contains two main ideas: 1. the real reason behind the conception of open-plan offices. 2. the reason why companies put this idea into practice. Option 1 is an inferential statement—the passage doesn't state that the workers were unsatisfied in the new setup. Option 2 is incorrect—the wall-free office plan did work out, though it served a different purpose. Reject it. Option 3 summarizes the passage correctly—it brings in both the major ideas stated above. Retain it. Option 4 distorts the information given in the passage—wall-free office spaces didn't quite work out as desired as it became tools for exploitation of labour; cubicles were an attempt to put some soul back in. In other words, cubicles replaced wall-free office spaces not because the former was better at breaking down the social walls. Reject it. Option 3 best summarizes the passage. Hence, [3].

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Q.34) Statement 1 states the primary reason why ocean plastic is problematic—its intake by marine animals. Sentence 2 gives us an idea of the particle size of the ocean plastic. Sentence 3 states another harmful effect of ocean plastic—absorption of pollutants from the water column. Sentence 4 classifies ocean plastic as 'one of the most important and pervasive environmental problems today'. Sentence 5 explains why intake of ocean plastic is harmful for marine animals. When rearranged, four of these sentences form a logical sequence—4153. Sentence 2, because it doesn't specify exactly why the small size of ocean plastic may be harmful, do not fit anywhere in the above sequence. Hence, [2].

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SECTION-II
DATA INTERPRETATION & LOGICAL REASONING

35. There are a total of 7 states which have actual rainfall greater than or equal to 900 mm.
Out of these only 3 have negative deviation. (Arunachal, Kerela and Meghalaya)
Required percentage = (3/7) × 100 ≈ 42.86%.
Hence option 4.

36. There are 9 states whose actual rainfall is less than or equal to 750 mm.
The states arranged according to the deviation is: Gujarat (25%), Karnataka(20%), Rajasthan(15%), MP(10%), Assam(−10%), WB(−30%), Jharkhand(−35%), Delhi(−40%) and Manipur(−60%).
So the median deviation is of Assam (−10%).
Hence option 3.

37. There are five states who have actual rainfall of 600 mm or less in 2019 and have a negative deviation from LPA [Assam(600). WB(600), Jharkhand (400), Manipur (400) and Delhi (300).

Required average = (600 + 600 + 400 + 400 + 300)/5 = 460 mm.

Hence option 1.

38. Actual rainfall in Gujarat in 2019 is 20% more than the rainfall in 2009.

So, considering Gujarat, Actual rainfall in 2019 = (6/5) × Actual rainfall in 2009 = 600 mm (given).

∴ Actual rainfall in 2009 = 500 mm.

As deviation for Gujarat is 25%, so average rainfall for 2009 to 2018 is 600/1.25 = 480 mm.

So, total rainfall from 2009 to 2018 = 480 × 10 = 4800 mm.
LPA for 2019 = (Rainfall in 2010 to 2019)/10 = [(Rainfall in 2009 to 2018) − Rainfall in 2009 + Rainfall in 2019]/10
= (4800 − 500 + 600)/10 = 490 mm.

Hence option 3.

39. Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.
Therefore, Each of MT and ET had at least 4 + 3 + 2 = 9 questions. Also, these 9 questions were worth 80 marks. So, remaining 20 marks can be from (one 5-mark and one 15-mark questions OR two 10-mark questions OR two 5-mark questions and one 10-mark question OR four 5-mark questions).
From (ii), Annie prepared exactly one question (i.e., fifth question) for MT and minimum number of questions in MT = 1 + 5 × 5 = 11
From (iv) and (v), Fakir and Chetan prepared the first and the third questions respectively. Now from (ii) and (iii), Fakir and Chetan prepared the second and the fourth questions also.

Similarly it can be concluded that Dave prepared first two questions of ET and Chetan prepared the third and the fourth questions of ET.
Also, as each faculty member prepared the same number of questions and Chetan prepared 4 questions, each prepared 4 questions.
Total number of questions in ET and MT together = 4 × 6 = 24. As ET contained more questions than MT, MT had 11 questions and ET had 13 questions.
Therefore, Beti (6th and 7th), Esha (8th and 9th) and Dave (10th and 11th) prepared exactly two questions for MT. Also, first five questions were of 5 marks, then three 10-mark questions and in the last, three 15-mark questions. As ET has 13 questions and 9 among these contribute to 80 marks, we are left with the possibility that there were four 5-mark questions other than the 9 questions.
Annie prepared 3 questions for ET (i.e., the 5th, 6th and 7th). Esha prepared 8th and 9th questions. And Fakir prepared last two questions. Thus, we have

Q. No.	ET	MT
1	Dave(5-mark)	Fakir(5-mark)
2	Dave(5-mark)	Fakir(5-mark)
3	Chetan(5-mark)	Chetan(5-mark)
4	Chetan(5-mark)	Chetan(5-mark)
5	Annie(5-mark)	Annie(5-mark)
6	Annie(5-mark)	Beti(10-mark)
7	Annie(5-mark)	Beti(10-mark)
8	Esha(5-mark)	Esha(10-mark)
9	Esha(10-mark)	Esha(15-mark)
10	Beti(10-mark)	Dave(15-mark)
11	Beti(10-mark)	Dave(15-mark)
12	Fakir(15-mark)	
13	Fakir(15-mark)	

So, The second question in ET was prepared by Dave.

Hence, option 4.

40. Refer to the table obtained in the solution of the first question.
There were eight 5-mark questions in ET and five 5-mark questions in MT.
Thus, there were 13 (5-mark) questions in MT and ET combined.
Hence, option 1.

41. Refer to the table obtained in the solution of the first question.
Only Esha and Dave prepared 15-mark questions for MT while Fakir prepared 15-mark questions for ET.
Hence, option 1

42. Refer to the table obtained in the solution of the first question.
Beti prepared 10th and 11th question in ET.
Hence, option 1.

43. Consider C1R1: Maximum coins in a pouch = 4 and minimum coins = 2. The third bag could have 2, 3 or 4 coins. Therefore, sum = 8, 9 or 10

Consider C1R2: Maximum coins in a pouch = 5 and minimum coins = 3. The third bag could have 3, 4 or 5 coins. Therefore, sum = 11, 12 or 13
Consider C1 R3: Maximum coins in a pouch = 2 and minimum coins = 1. As the sum = 4, third pouch has 1 coin.
As the sum of coins in the nine pouches in the column are divisible by 9, the coins in C1R1 and C1R2 has to be 10 (i.e., 2, 4, 4) and 13 (i.e., 3, 5, 5) respectively.
Consider column 2:
C2R1: Maximum coins in a pouch = 8 and minimum coins = 6. The third bag could have 6, 7 or 8 coins. Therefore, sum = 20, 21 or 22
C2R2: All the three pouches have one coin each. Sum = 3
C2R3: Maximum coins in a pouch = 2 and minimum coins = 1. The third bag could have 1 or 2 coins. Therefore, sum = 1 or 4
In order to have number of coins in the cells of the column divisible by 9, sum of the coins in C2R1 = 20(i.e., 6, 6, 8) and in C2R3 = 4(i.e., 1,1,2)

Now consider R1

First two cells together have 30 coins. So the third cell has to have 6(i.e., 1, 2, 3) coins.

Consider R2: First two cells together have 16 coins. Coins in the third cell are in the range 6 + 6 + 20 = 32 to 6 + 20 + 20 = 46. Therefore the third cell has to have 38 (i.e., 6, 12, 20) coins.

Consider R3: First two cells together have 8 coins. So the third cell has to have 10 (i.e., 2, 3, 5) coins.

Thus, we have

	Column 1	Column 2	Column 3
Row 1	(2, 4, 4)	(6, 6, 8)	(1, 2, 3)
Row 2	(3, 5, 5)	(1, 1, 1)	(6, 12, 20)
Row 3	(1, 1, 2)	(1, 1, 2)	(2, 3, 5)

First column of the second row has 13 coins. i.e., Rs. 13

Answer: **13**

44. Refer to the table obtained in the solution of the first question.

Two pouches in Row 3 Column 1 slot have 1 coin each.

Three pouches in Row 2 Column 2 have 1 coin each.

Two pouches in Row 3 Column 3 slot have 1 coin each.

One pouch in Row 1 Column 3 slot has 1 coin.

Total number of pouches = 2 + 3 + 2 + 1 = 8

Answer: **8**

45. Refer to the table obtained in the solution of the first question.

For average amount to be an integer, we need to consider slots having total number of coins in the pouches divisible by 3.

There are two such slot i.e., (row 1 column 3) and (row 2 column 2).

Answer: **2.**

46. The slots for which the total amount in its three pouches strictly exceeds Rs. 10 are Row 1 Column 2, Row 2 Column 1, Row 2 Column 3.

Answer: **3.**

47. Robert is in Team 1 and speaks Arabic and French.

So from (1) and (2); Team 1 speaks English, Chinese, Arabic and French.

So, Sally cannot be in Team 1. As Arabic is spoken by exactly one team, Robert is only on Team 1.

Therefore from (4), Paula and Sally are in Teams 2 and 3.

Quentin is in Team 3 and speaks Dutch and English.

So from (1) and (2); Team 3 speaks English, Chinese, Dutch and Basque. Also, Paula, Sally and Quentin are members of Team 3.

As Dutch is spoken by exactly one team, Quentin is only on Team 3. Team 3 has exactly 3 members.

Therefore from (1), each team has 3 members. So, Terence is the third member of Team 1 and 2.

Therefore, we have

Teams	Languages	Members
Team 1	English, Chinese, French, Arabic	Paula, Terence, Robert
Team 2	English, Chinese, French, Basque	Paula, Sally, Terence
Team 3	English, Chinese, Dutch, Basque	Paula, Sally, Quentin

So, Quentin is not a member of Team 2.

Hence, option 1.

48. Refer to the table obtained in the solution of the first question.

Quentin and Robert are members of exactly one team while Paula is a member of all the three teams. Sally is a part of exactly two teams.

Hence, option 1.

49. Refer to the table obtained in the solution of the first question.

Paula is a member of all teams.

Hence, option 2.

50. Refer to the table obtained in the solution of the first question.

Apart from Chinese and English, Team 1 speaks Arabic and French.

Hence, option 2.

51. The percentage data can be tabulated as follows.

	Revenue %			Cost %		
Department	**2016**	**2017**	**2018**	**2016**	**2017**	**2018**
Electronics	50	30	20	30	40	30
Clothing	20	30	40	30	30	20
Produce	30	40	40	40	30	50

Let the total cost of the store in 2016 be 100.

From (1): Total revenue in 2016 = 2 × Total cost in 2016 = 2 × 100 = 200.

From (2): Total revenue in 2017 = 2 × Total revenue in 2016 = 2 × 200 = 400.

From (2): Total cost in 2018 = 2 × Total cost in 2016 = 2 × 100 = 200.

So, we can find the department wise revenue in 2016 and 2017 and cost in 2016 and 2018 using the percentages found from the radar graph.

	Revenue			Cost			Profit		
Department	**2016**	**2017**	**2018**	**2016**	**2017**	**2018**	**2016**	**2017**	**2018**
Electronics	100	120		30		60	70		
Clothing	40	120		30		40	10		
Produce	60	160		40		100	20		
Total	200	400		100		200	100		

From (3): Cost from Electronics in 2017 = Revenue from Electronics in 2017 = 120.

So total cost in 2017 = 120/40% = 300. And hence the department wise costs can be found for 2017.

From (4): Revenue from Clothing department in 2018 = Cost for the Produce departement in 2018 = 100.

So total revenue in 2018 = 100/40% = 250.

	Revenue			Cost			Profit		
Department	2016	2017	2018	2016	2017	2018	2016	2017	2018
Electronics	100	120	50	30	120	60	70	0	-10
Clothing	40	120	100	30	90	40	10	30	60
Produce	60	160	100	40	90	100	20	70	0
Total	200	400	250	100	300	200	100	100	50

Revenue generated from the Produce department in 2017 to that in 2018 = 160 : 100 = 8 : 5.

Hence option 2.

52. Refer to the final table obtained in the solution of the first question.

Required percentage = (Profit/Cost) × 100 = (50/200) × 100 = 25%.

Answer: **25.**

53. Refer to the final table obtained in the solution of the first question.

Total profit in 2016 = 100.

Profit from Electronics in 2016 = 70/

Required percentage = (70/100) × 100 = 70%.

Answer: **70.**

54. Refer to the final table obtained in the solution of the first question.

Profit percentage of the store;

In 2017 = (100/300) × 100 = 33.33%.

In 2018 = (50/200) × 100 = 25%.

Required difference = 33.33 − 25 = 8.33 ≈ 8.3.

Hence option 3.

55. As the top three performances in each round were awarded 7, 3 and 1 respectively, remaining all performances were awarded 0 points.

Amita was participant in round 1 and in rounds 6 through 10.

So, in all she scored 8(= 1 + 7) points in round 1 and round 6. Also she scored 10 points in rounds 7 through 10 together.

Bala was participant in rounds 1 and 2 and also in rounds 7 through 10.

As he **scored 2 points in first two rounds, he must have scored 1 point in each of the two rounds**. He scored 3(= 3 or 1+1+1) points in rounds 7 through 10.

Chen scored 3 points in first three rounds and as he scored in three consecutive rounds among 7 through 10, he must have scored **1 point each in round 8, 9 and 10.**

David scored 6 points in first four rounds together and **0 (i.e., 6 – 6) each in 9th and 10th rounds.**

Eric scored 3(= 3 or 1+1+1) points in five rounds together and hence 10 – 3 = **7 points in the 10th round.**

Gordon scored 17 points in rounds 2 through 6 together and hence, 17 – 17 = **0 points in the 7th round.**

As he did not scored consecutively in any two rounds, **he must have scored 0 in 3rd and 5th round** and 7, 7, and 3 points in 2nd, 4th and 6th rounds, not necessarily in the given sequence.

Hansa scored 1 point in one of the rounds 3-6. So, she must have scored 4 – 1 = 3 points in either 7th or 8th rounds.

Ikea scored 2 points in 4th, 5th and 6th rounds together. Therefore, remaining 15 points must have been scored in 7th, 8th and 9th round in some order.

As Chen scored 1 point in the 8th and 9th round, **Ikea must have scored 1 point in 7th round and 7 points each in the 8th and 9th round.**

Joshin scored 14 points in 5th and 6th rounds. So he scored 7 points **in each of 5th and 6th rounds.**

Now we need to decide about remaining 3 points in the last four rounds.

As Joshin scored in round 7, and Chen has scored 1 point in each of the last three rounds, Joshin must have scored all **3 points in the 7th round and 0 points in the last three rounds.**

Therefore, we can tabulate this information as

Player No.	Player Name	1	2	3	4	5	6	7	8	9	10
1	Amita		NA	NA	NA	NA					
2	Bala	1	1	NA	NA	NA	NA				
3	Chen				NA	NA	NA	NA	1	1	1
4	David					NA	NA	NA	NA	9	0
5	Eric						NA	NA	NA	NA	7
6	Fatima							NA	NA	NA	NA
7	Gordon	NA		0		0		0	NA	NA	NA
8	Hansa	NA	NA							NA	NA
9	Ikea	NA	NA	NA				1	7	7	NA
10	Joshin	NA	NA	NA	NA	7	7	3	0	0	0

As Joshin scored 3 points in the 7th round, Hansa must have scored 3 in 8th round and 0 in the 7th round. Now consider 7th round. Only Amita must have scored 7 in the 7th round.

As she has scored in round 10, the remaining 3 points she must have scored in round 10. Thus, we can now conclude that Bala scored 3 points in the 9th round.

As Joshin scored 7 points in the 6th round Amita must have scored 1 point in this round and hence7 points in the first round.

Gordon scored 7, 7 and 3 points in 2nd, 4th and 6th rounds. Joshin scored 7 points in the 6th round so, Gordon scored 7, 7 and 3 points in 2nd, 4th and 6th rounds respectively.

So we can fill remaining values as 0 in the table for the 6th round.

Thus, Ikea scored 1 point each in 4th and 5th rounds. And hence, Hansa scored 1 point in the 3rd round.

So far we have entered values 1 and 7 for all the rounds. Now we have to enter 3 points of each of Bala, Chen, David and Fatima and 7 points for Fatima.

Eric and Fatima both scored in a round. This is possible only when Fatima scored 7 points and Eric scored 3 points. This must be round 3.

No one other than Fatima scored 3 points in round 5. Similarly, David scored 3 points in the 4th round. Chen and David scored 3 points in 1st and 2nd round in some order.

Thus we have

Player No.	Player Name	1	2	3	4	5	6	7	8	9	10
1	Amita	7	NA	NA	NA	NA	1	7	0	0	3
2	Bala	1	1	NA	NA	NA	NA	0	0	3	0
3	Chen	0/3	3/0	0	NA	NA	NA	NA	1	1	1
4	David	3/0	0/3	0	3	NA	NA	NA	NA	9	0
5	Eric	0	0	3	0	0	NA	NA	NA	NA	7
6	Fatima	0	0	7	0	3	0	NA	NA	NA	NA
7	Gordon	NA	7	0	7	0	3	0	NA	NA	NA
8	Hansa	NA	NA	1	0	0	0	0	3	NA	NA
9	Ikea	NA	NA	NA	1	1	0	1	7	7	NA
10	Joshin	NA	NA	NA	NA	7	7	3	0	0	0

After round 3, the scores of Chen, David and Eric were 3, 3, 3.

Hence, option 3.

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56. Refer to the table obtained in the solution of the first question.

The three players in the last position after Round 4 were Hansa, Ikea and Joshin.

Hence, option 4.

57. Refer to the table obtained in the solution of the first question.

Ikea scored in maximum number of rounds i.e., in 5 rounds.

Hence, option 1.

58. Refer to the table obtained in the solution of the first question.

Amita, Chen and Eric scored in the last round.

Hence, option 3.

59.

	Sunita	Ragini
Only proposal A	k	p
Only proposal B	l	q
Both A and B	m	r
Neither A nor B	n	s

From (1), k + m + p + r = 250 and l + m + q + r = 250

From (4), l + m = 0.2 × 250 = 50

From (6), **l = 0 ⇒ m = 50**

∴ q + r = 200

From (2), k + l + m + n = 200 and hence p + q + r + s = 300

From (3), p + r = 0.3 × 300 = 90 ⇒ k + m = 160

As l = 0 and k + l + m + n = 200, **n = 40**

∴ **k = 110**

From (7), n + s = 100 ⇒ **s = 60**

∴ p + q + r = 240 and p + r = 90 ⇒ q = 150

∴ q + r = 200 ⇒ r = 50 and hence p = 40

Thus, we have

	Sunita	Ragini
Only proposal A	110	40
Only proposal B	0	150
Both A and B	50	50
Neither A nor B	40	60

Out of 100 students who supported both proposals A and B, 50 students preferred Sunita as student union president. i.e., 50%

Hence, option 1

60. Refer to the final table obtained in the solution of the first question.

Among 250 students who supported proposal A, 110 + 50 = 160 students preferred Sunita for student union president which is equivalent to 64%.

Answer: **64.**

61. Refer to the final table obtained in the solution of the first question.

250 students did not support proposal A. Out of these, 150 + 60 = 210 students preferred Ragini as student union president, which is equivalent to 84%.

Answer: **84.**

62. Refer to the final table obtained in the solution of the first question.

Out of 250 students who supported proposal B, 150 did not support proposal A and preferred Ragini as student union president.

Hence, option 3.

63. The clinic is open from 9 a.m. to 11:30 a.m. i.e., for 150 minutes.

In these 150 minutes, Dr. Ben can see 15 patients, Dr Kale can see 10 patients and Dr Wayne can see 6 patients.

Thus, the clinic can cater to 15 + 10 + 6 = 31 patients.

Hence, option 1.

64. The clinic is open from 9 a.m. to 11:30 a.m. i.e., for 150 minutes.

As the queue is never empty, in these 150 minutes, Dr. Ben can see 15 patients, Dr Kale can see 10 patients and Dr Wayne can see 6 patients and thereby earn Rs. 1,500, Rs. 2,000 and Rs. 1,800 respectively.

i.e. Dr. Kane earns the maximum amount in consultation charges on that day.

Hence, option 2.

65. In first 50 minutes, Dr. Ben sees 5patients; Dr. Kane sees 3 patients and he must be with his four patients while Dr. Wayne sees 2 patients.

These must be first 11 patients. Now about patients with token numbers 12 and 13.

On Monday patient with token number 13 will go to Dr. Wayne, on Wednesday and Friday he will go to Dr. Ben.

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Hence, on Monday, he will spend 10 + 50 + 25 = 85 minutes, while on the remaining two days, he will spend 10 + 50 + 10 = 70 minutes.

So, Mr. Singh was at the clinic for the maximum duration on Monday.

Hence, option 2.

66. Patient with token no. 1 will be seen by Dr. Wayne for 25 minutes and that with token number 2 will be seen by Dr. Ben for 10 minutes.

Now the two patients those entered at 9:15 a.m were given tokens with numbers 3 and 4.

These will be seen by Dr. Ben and Dr. Kane respectively.

As Dr. Kane takes 15 minutes to examine one patient, he will be free at 9:30 am.

Thus, in first half an hour three doctors will not be simultaneously free.

This is true for duration of 30 minutes henceforth.

Thus the total duration in minutes when all three doctors are simultaneously free will be 0.

Hence, option 1.

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SECTION-III
QUANTITATIVE ABILITY

67. Let the purchase price of one bicycle be $100x$.

$\therefore$ Total purchase price for all 10 bicycles= $1000x$

Total selling price of six bicycles (at 25% profit) = $6 \times 125x = 750x$

Total selling price of the remaining four bicycles (at 25% loss) = $4 \times 75x = 300x$

$\therefore$ Total selling price of all the ten bicycles = $750x + 300x = 1050x$

Total profit = Total selling price − Total purchase price = 2000 (given)

$= 1050x - 1000x = 50x = 2000$

$\therefore x = 40$.

So purchase price of one bicycle = $40 \times 100 = 4000$.

Hence, option 1.

68. For a quadratic equation $ax^2 + bx + c$ to have real and distinct roots, its discriminant needs to be greater than zero.

$\therefore b^2 - 4ac > 0$

$\therefore 4^2 - [4 \times 1 \times (-\log_2 A)] > 0$

$\therefore 16 > -4 \log_2 A$

$\therefore 4 > -\log_2 A$

$\therefore \log_2 A > -4$

$\therefore A > 2^{-4}$

$\therefore A > 1/16$

Hence, option 1.

69. Every minute a motor cycle leaves A and moves towards B. The last motor cycle (45th) reaches B at 11am.

The motorcycles start from A at 10: 01, 10:02 ...and so on. So we can infer that the last motorcycle (45th) would have left A at 10:45 and reaches B at 11 am.

Since the speed of each motorcycles is same, hence each motorcycle takes 15 minutes (10:45 am to 11 am) to reach B from A.

When the cyclist doubles its speed, it will reach in half the time i.e. 30 minutes. So the cyclist reaches B at 10:30 am.

In this case, the last motorcycle to reach B would have to start at 10:15 am. So this must be the 15th motorcycle.

∴ Total 15 motorcycles would have started from A and reached B by the time the cyclist reached B.

Hence, option 1.

70. 1, 3, 5 .... 47 constitutes an AP.

So, $47 = 1 + (k - 1) \times 2$ (where k is the number of terms in this AP)

$\therefore k = 24$.

Now the expression can be written as $(24 \times 2n) + (1 + 3 + 5 + 7 + ... + 47)$

$= 48n + (24/2)[2(1) + \{(24 - 1) \times 2\}]$

$= 48n + 576 = 5280$ (given)

$\therefore n = 98$

$\therefore 1 + 2 + 3 + 4 + ... + 98 = (98 \times 99)/2 = 49 \times 99 = 4851$

Answer: **4851.**

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71. The expressions can be written as:

$[(n + 4)(n + 3)/(n - 4)(n + 3)]$

$= (n + 4)/(n - 4)$

$= (n - 4 + 8)/(n - 4)$
$= 1 + [8/(n - 4)]$

For this expression to be a positive integer, $(n - 4)$ needs to be a factor of 8.

The various values of $(n - 4)$ are 1, 2, 4 and 8.

So, $n = 5, 6, 8$ and 12.

$\therefore n_{max} = 12$.

Hence, option 4.

72. Let 2^{3x} be t.

We can rewrite the expression as, $t^2 + 4t - 21 = 0$

$= (t + 7)(t - 3) = 0$

$\therefore t = -7, 3$

$t = 2^{3x}$, so t cannot be negative.

$\therefore 2^{3x} = 3$

Taking log both sides, we get;

$3x \log 2 = \log 3$

So, $x = (\log_2 3)/3$

Hence, option 4.

73. The expression can be written as :

$= 2^8 \times 3^5 \times 5^4$

The factors which are perfect squares will contain the following factors : $(2^0, 2^2, 2^4, 2^6, 2^8)$, $(3^0, 3^2, 3^4)$ and $(5^0, 5^2, 5^4)$

Total number of factors = $5 \times 3 \times 3 = 45$

One of these factors is 1.

Required number of factors = $45 - 1 = 44$.

Answer: **44.**

74. Let the scores of Rama, Mohan and Anjali be R, M and A respectively.

Given: $(A + 6) : (M + 6) : (R + 6) = 11 : 10 : 3$.

$(A + 6) - (R + 6) = (A - R) = (11 - 3)k = 8k$.

So the answer has to be multiple of 4, so options 1 and 4 are discarded.

Initially, $12R = M + A$

$\therefore 12(3k - 6) = (10k - 6) + (11k - 6)$

$\therefore k = 4$.

So, $(A - R) = 8k = 8 \times 4 = 32$.

Hence option 2.

75. Let the salaries of Ramesh, Ganesh and Rajesh in 2010 be 60, 50 and 70 respectively.

If Ramesh's salary increses by 25%, so his salary in 2015 = 60 × (5/4) = 75.

The ratio of the salaries of Ramesh, Ganesh and Rajesh in 2015 is 3 : 4 : 3. Let these salaries be $3x$, $4x$ and $3x$ respectively.

$\therefore 3x = 75 \Rightarrow x = 25$.

∴ Salaries of Ramesh, Ganesh and Rajesh in 2015 are 75, 100 and 75 respectively.

∴ Increase in salary of Rajesh during the period = 75− 70 = 5.

Required percentage = (5/70) × 100 = 50/7 = 7.1428 ≈ 7%.

Hence, option 2.

76. In the expression we put;

$n = 1$ to get; $\boldsymbol{a_1 = 1}$ ...(I)

$n = 2$ to get; $a_1 - a_2 = 2$ ...(II)

$\therefore \boldsymbol{a_2 = -1}$.

$n = 3$ to get; $a_1 - a_2 + a_3 = 3$ ...(II)

$\therefore \boldsymbol{a_3 = 1}$.

$n = 4$ to get; $a_1 - a_2 + a_3 - a_4 = 4$ ...(II)

$\therefore \boldsymbol{a_4 = -1}$.

So the pattern followed is 1, −1, 1, −1 and so on.

So $a_{odd} = 1$ and $a_{even} = -1$.

51, 52, 53 .... 1023 is an AP with number of terms = [(1023 − 51)/1] + 1 = 973.

This series has 486 even terms and 487 odd terms.

$\therefore a_{51} + a_{52} + \ldots + a_{1023} = (487 \times 1) + (486 \times -1) = 1.$

Hence option 4.

77. Let's say John does 'a' hours of regular work and 'b' hours of overtime work.

$a + b = 172$ ....(I)

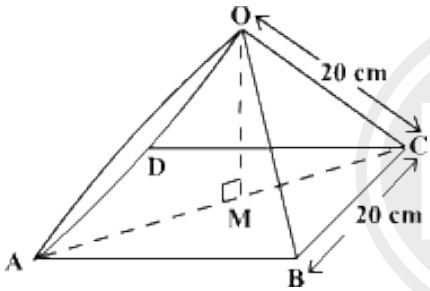
Also, $(\text{Income})_{\text{Overtime}} = 15\%$ of $(\text{Income})_{\text{Regular}}$.

$\therefore 114b = 0.15 \times 57a$ ...(II)

Solving (I) and (II), we get; $b = 12$ and $a = 160$.

Answer: **12.**

78.



In the above image, the side length of the square is equal to the side length of the equilateral triangle.

So, AB = BC = OC = 20 cm.

ΔAOM is a right angled triangle with ∠AMO = 90°, AO = 20 cm. OM is the height of the pyramid.

AC is the diagonal of the square base, so AM = AC/2 = (20√2)/2 = 10√2.

So using pythagoras theorem; $OM^2 = AO^2 - AM^2 = 20^2 - (10\sqrt{2})^2 = 400 - 200 = 200$.

∴ OM = 10√2.

Hence, option 2

79. Interior angle of a regular polygon having n sides $A_n = [(n - 2)/n] \times 180$

Give that $A_b = (3/2) \times A_a$.

$\therefore [(b - 2)/b] \times 180 = (3/2) \times [(a - 2)/a] \times 180$

Use $b = 2a$ to get; $a = 4$ and $b = 8$.

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$a + b = 4 + 8 = 12$.

Each interior angle of a regular polygon having 12 sides = $[(12 - 2)/12] \times 180 = 150°$.

Answer: **150.**

Alternatively;

Most questions of CAT can be solved even if you dont remember the relevant formula. But the time taken to solve the question is lesser if you remember the formula. So although you are encouraged to memorize the formula, you can arrive at the answer logically as shown below.

Let the number of sides of the two regular polygons be a and $2a$. Let each interior angle be x and $1.5x$ respetively.

Sum of all interior angles of the regular polygon with;

a sides = xa.

$2a$ sides = $(1.5x)(2a) = 3xa$.

So number of sides double whereas the sum of interior angles becomes 3 times.

Now we make a table where we write the number of sides in a regular polygon and the respective sum of interior angles. Note that this sum is always a multiple of 180.

n	Sum of interior angles S	
3	180×1	180
4	180×2	360
5	180×3	540
6	180×4	720
7	180×5	900
8	180×6	1080

If we take n as 3 and 6, $(6 = 2 \times 3)$: The ratio of the angles is not 1 : 3. (As 180 : 720 = 1 : 4). So this case is invalid.

If we take n as 4 and 8, $(8 = 2 \times 4)$: The ratio of the angles is 360 : 1080 = 1 : 3. So this case is valid.

Since this is a numerical entry question, so there will only one answer and there is no need to check for further cases.

So, sum of Sum of all interior angles of the regular polygon with 8 + 4 = 12 sides = $180 \times 10 = 1800$.

Required interior angle = 1800/12 = 150.

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Answer: **150.**

80. Let the meeting point be Q. When A and B are at Q, A and B have covered 60% and 40% of the track respectively.

∴ Ratio of speeds of A and B = 60 : 30 = 3 : 2.

A takes 12 minutes (to cover 40% of the distance) to go from Q to P at a speed of 3*x* (say).
So, time taken by A (to cover 60% of the distance) to go from P to Q = (12/40) × 60 = 18 minutes.

Time taken by B at a speed of 2*x* (to cover 60% of the distance) to go from Q to P = 18 × (3/2) = 27 minutes.

So, B will return to P at 10:27 am.

Hence, option 2.

81. Both the sequences are APs. So, if we only consider the common terms, that sequence will also be in AP with the common difference *d* equal to the L.C.M of the common differences of the two given sequences.

∴ d = L.C.M (4 , 5) = 20.

Consider the sequence consisting of common terms: The general term $T_n = 19 + (n - 1)d$ and the greatest number in this series is less than or equal to 415. (415 is the smaller of the two between 415 and 464)

∴ $T_n = 19 + [(n - 1) \times 20] = 20n - 1$.

∴ $20n - 1 \leq 415$

∴ $n \leq 20.8$.

So, $n_{max} = 20$.

So, there are 20 common terms.

Hence, option 2.

82. Total Amount received by Amal = [(12000 × 1.08) + (10000 × 1.03 × 1.03)] = (12960 + 10609) = Rs. 23569

∴ Total interest received by him = 23569 − 22000 = Rs. 1569

Since, Amal and Bimal receive the same amount of interest, Bimal also recieves Rs 1569 as interest.

∴ Priniciple amount invested by Bimal at 7.5% simple interest for one year = (1569/7.5)× 100 = Rs. 20920.

Answer: **20920.**

83. Ratio of speeds of John and Mary = 6 : 7.5 = 4 : 5. So let speeds of Johan and Mary be $4k$ and $5k$ respectively.

Let length of tracks A and B be a and b respectively.
Tiime taken by John to make 9 rounds of track A = Time taken by Mary to make 5 rounds of B.

$\therefore 9a/4k = 5b/5k$.

$\therefore b = 2.25a$ ....(I)

$a + b = 325$ ...(II)

$\therefore a = 100$ m and $b = 225$ m.

So, $a = 0.1$ km.

$\therefore$ Time taken by Mary to complete one round of A = $a/7.5 = 0.1/7.5 = 0.01333$ hrs = 48 seconds.

Answer: **48.**

84. The given equation can be rewritten as $n^2 - m^2 = 105$.

$\therefore (n + m)(n - m) = 105$.

Since n and m are positive, $(n + m)$ is also positive and hence from equation (I), we can deduce that $(n - m)$ is positive.

So, $(n - m) > 0$ or $n > m$.

There are **8 factors** of 105: 1, 3, 5, 7, 15, 21, 35 and 105.

Both m and n are positive integers, so $(n + m) > (n - m)$.

$\therefore (n + m)(n - m) = (105 \times 1)$ or (35×3) or (21×5) or (15×7)

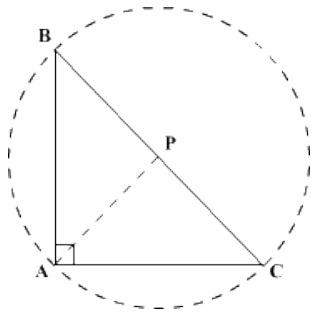
Let $(n + m)$ be a and $(n - m)$ be b. So, $2n = (a + b) \Rightarrow n = (a + b)/2$.

So, $(a + b)$ needs to be divisible by 2 which is true for all four cases [$(a, b) = (105, 1)$ or $(35, 3)$ or $(21, 5)$ or $(15, 7)$].

So, possible number of ordered pairs (m, n) is 4.

Answer: **4.**

85.



∠BAC = 90°, so BC can be considered diameter of a circle with center anywhere on BC.

AP is maximum when it is the perpendicular bisector from the vertex to the hypotenuse. So, BP = PC = 20/2 = 10. This implies that P is the center of the circle.

∴ AP = BP = PC = 10 cm.

Hence, option 2

86. Let the six digit number be *abcdef*.

$f = a + b + c$

$e = a + b$

$c = a$

$b = 2a$

$d = e + f$

Let, $a = k$.

∴ $b = 2k$; $c = k$; $f = a + b + c = 4k$; $e = 3k$ and $d = e + f = 7k$.

Fourth digit = $7k$. It can take a maximum value of 7 when $k = 1$.

Answer: **7.**

87. Let the total job to be done be the a multiple of the LCM of 40 and 20 which is 40 units.

Work done in a day;

By Anil = 40/20 = 2 units.

By Sunil = 40/40 = 1 unit.

In the first 3 days, amount of work done by Anil = 2 × 3 = 6 units.

Amount of work remaining = 34 units.

Since Bimal does 10% of the job (= 4 units), the remaining 30 units (34 − 4) has to be done by Anil and Sunil

They together do (2 + 1) = 3 units of work in a day

∴ Time taken by them to do 30 units of work = 30/3 = 10 days.

So, total number of days = [10 (for 34 units work) + 3 (for 6 units work)] = 13 days.

Hence, option 1.

88. Units digit of any power of 5 is 5.

Units digit of any power of 3 is 3 or 9 or 7 or 1.

$5^{x-1} + 3^{y+1} = 9686$. Considering only the units digits: 5 + Units digit of $3^{y+1} = 6$.

∴ Units digit of $3^{y+1} = 1$.

This is possible only when $(y + 1)$ is multiple of 4.

i.e. $(y + 1) = 4, 8, 12, ..$ and so on.

Also, $3^{y+1} < 9686$

Note that $3^{12} = (3^4)^3 > 9676$

∴ $(y + 1) \neq 12$

If $(y + 1) = 4$, $3^{y+1} = 81$ and hence $5^{x-1} + 81 = 9686 \Rightarrow 5^{x-1} = 9605$ which is not possible.

If $(y + 1) = 8$, $3^{y+1} = 6561$ and hence, $5^{x-1} + 6561 = 9686 \Rightarrow 5^{x-1} = 3125$ i.e. $(x - 1) = 5$.

∴ $x = 6$ and $y = 7$.

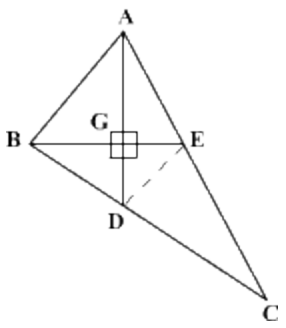
It can be cross checked with the other equation.

LHS = $5^x - 3^y = 5^6 - 3^7 = 13438$ = RHS.

∴ $x + y = 6 + 7 = 13$.

Answer: **13.**

89.



As AD and BE are medians, so G is the centroid of ΔABC.

So, AG : GD = BG : GE = 2 : 1.

∴ AG = 8 cm, GD = 4 cm, BG = 6 cm and GE = 3 cm.

CD/CB = CE/CA = 1/2.

∴ Area ΔEDC/ Area ΔABC = $(1/2)^2$ = 1/4. So let area of ΔEDC and ΔABC be a and $4a$.

So, A(□ABDE) = $4a - a = 3a$.

Also, A(□ABDE) = A(ΔABD) + A(ΔADE) = [(1/2) × 12 × 6] + [(1/2) × 12 × 3] = 54 sq. cm.

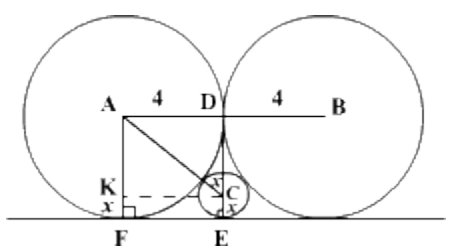
∴ $3a = 54$.

∴ $a = 18$.

So, area (ΔABC) = $4a = 4 \times 18 = 72$ sq. cm.

Hence option 1.

90. Let the radius of the third circle be 'x' cm.



AF = DE = 4 cm ⇒ AK = DC = $(4 - x)$ cm

In ΔADC, $AC^2 = AD^2 + DC^2$.

$(4 + x)^2 = 4^2 + (4 - x)^2$

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$\therefore x = 1.$

Hence option 1.

91. $\sqrt{\log_e [(4x - x^2)/3]}$ is a real number.

$\therefore \log_e [(4x - x^2)/3] \geq 0$

$\therefore (4x - x^2)/3 \geq e^0$

$\therefore 4x - x^2 \geq 3$

$\therefore x^2 - 4x + 3 \leq 0$

$\therefore (x - 3)(x - 1) \leq 0$

$\therefore 1 \leq x \leq 3.$

Hence option 1.

92. $f(2) = f(1 \times 2) = f(1)\, f(2)$

$\therefore f(1) = 1$ as $f(2) \neq 0$.

Also $f(1) < f(2)$

Now, $f(4) = f(2 \times 2) = [f(2)]^2$

$f(8) = f(2 \times 4) = f(2) \times [f(2)]^2 = [f(2)]^3$

$f(24) = f(3 \times 8) = f(3) \times [f(2)]^3 = 54 = 2 \times 3^3$

$\therefore f(3) = 2$ and $f(2) = 3$.

$f(18) = f(2 \times 9) = f(2) \times [f(3)]^2 = 3 \times 2^2 = 12.$

Answer: **12.**

93. Let the number of fiction and non-fiction books in 2010 be f and n respectively.

$\therefore f + n = 11500$ ...(I)

$\therefore$ Number of fiction books in 2015 = $1.1f$.

Number of non-fiction books in 2015 = $1.12n$.

$\therefore 1.1f + 1.12n = 12760$ ...(II)

Solving (I) and (II), we get;

$f = 6000$.

∴ Number of fiction books in 2015 = 1.1 × 6000 = 6600.

Hence option 1.

94. Amount of salt in 100 ml of solution A= 10% of 100 = 10 gm.

After the first operation, salt in vessel B = (22% of 500) + 10 = 110 + 10 = 120 gm in 600 ml solution.

Also vessel A will now have 400 ml solution with 40 gm salt in it.

Amount of salt in 100 ml solution from vessel B = 120/6 = 20 gm.

After adding this 100 ml solution from vessel B to vessel C.

Amount of salt in vessel C = (32% of 500) + 20 = 160 + 20 = 180 gm in 600 ml solution.

Amount of salt in 100 ml of the solution in C = 180/6 = 30 gm.

Finally when this 100 ml is transferred to vessel A, 500 ml of solution in vessel A will have 40 + 30 = 70 gm of salt.

Required percentage = (70/500) × 100 = 14%.

Hence option 1.

95. Average of 30 integers is 5 ⇒ Sum of the 30 integers = 150

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Since, we need to maximise the average of given 20 integers, we need to minimise the other 10 integers

These 10 integers can take a minimum value of 6, so their minimum sum = 10 × 6 = 60.

Maximum sum of other 20 integers = 150 − 60 = 90.

Average of these 20 integers = 90/20 = 4.5.

Hence, option 4.

96. As the total number of cylinders to be kept at a minimum, the volume of each cylinder must be maximum possible.

HCF(405, 783, 351) = 27 ⇒ Volume of each cylinder = 27 cc.

∴ $\pi r^2 h = 27$ (where r and h represent the radius and height of the cylinder)

Given that $r = 3$, so $\pi h = 3$.

Number of cylinders of;

Iron = 405/27 = 15.

Aluminium = 783/27 = 29.

Copper = 351/27 = 13.

∴ Total number of cylinders = 15 + 29 + 13 = 57.

Total surface area of all the cylinders = $57 \times [2\pi r^2 + 2\pi rh]$

Put $r = 3$ and $\pi h = 3$ to get;

Required surface area = $1026(1 + \pi)$ sq. cm.

Hence, option 4.

97. A = (9/10) B; B = (5/4) C and C = (4/5) D.

∴ B = (10/9)A = (10/9) × 72 = 80.
∴ C = (4/5) B = (4/5) × 80 = 64.

∴ D = (5/4) C = (5/4) × 64 = 80.

Answer: **80.**

98. $(ax + by) = 65$

Squaring both sides we get;

$(ax + by)^2 = a^2x^2 + b^2y^2 + 2abxy = 4225$ ...(I)

$a^2 + b^2 = 25$, $x^2 + y^2 = 169$.

Multipling the above two expressions we get;

$(a^2 + b^2)(x^2 + y^2) = 25 \times 169 = 4225$

$\therefore a^2x^2 + a^2y^2 + b^2x^2 + b^2y^2 = 4225$ ...(II)

Equating (I) and (II), we get;

$a^2y^2 + b^2x^2 = 2abxy$

$\therefore a^2y^2 + b^2x^2 - 2abxy = 0$

$\therefore (ay - bx)^2 = 0$

$\therefore k = ay - bx = 0.$

Hence option 2.

99. In a quadratic equation, $ax^2 + bx + c = 0$, sum of roots = $-(b/a)$ and product of roots = (c/a).
For the given quadratic equation, $a = 1$.
$\therefore$ Sum of roots = $-b$ and Product of roots = c.
The two roots are $3a$ and $4a$.
$\therefore b = -7a$ and $c = 12a^2$
$\therefore b^2 + c = 49a^2 + 12a^2 = 61a^2$
Since, a is an integer, the answer should be a multiple of 61 and also a multiple of square of an integer.
The only option that satisfies is $61 \times 9 = 549$.
Hence, option 4.

100. Cost price to Amal = $1.2p \Rightarrow$ Cost price to Bimal $x = 1.3 \times 1.2p = 1.56p = x$.
Cost price to Asim = $0.8p \Rightarrow$ Cost price to Barun $y = 0.7 \times 0.8p = 0.56p = y$.
$(x - y)/p = (1.56p - 0.56p)/p = 1$.
Hence, option 3.

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